

Asset Tokenization: A Simulation Study

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Abstract

Asset exchanges (digital or real-world) seem to be one area that can benefit from blockchain technology. This means blockchain can be used to facilitate asset exchanges between parties without the need for an intermediary. In this research we look at how tokens issued on a platform (blockchain) can be used to represent and transfer ownership of an asset. This paper also looks into the important aspects to consider when tokenizing assets that may lead to the adoption of the process (i.e., parties choose blockchain as the primary option to exchange/ transfer ownership of assets). Thematic analysis is instrumental in allowing us to code the data that was found – related to asset tokenization. Thus, relevant themes (aspects) are derived to give a high-level view of the process of asset tokenization. We have created a model to give a bird's eye view of tokenization as a whole; from this, we derived the critical success factors for the adoption of asset tokenization.

Keywords

Keywords are not given

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Introduction

Blockchain is a very new technology that has been practically implemented and used for about a decade. It has a lot of potential to support many use cases other than just allowing payments to take place (i.e. blockchain was successfully implemented for the first time around 2009 as a peer to peer digital cash system that allowed the transfer of payment/ bitcoin from one peer to another (Larrier, 2021)). Tokenization seems to fall under one of the potential ways to use blockchain; and users of the technology stand a chance of being able to transfer assets in exchange for payment. The term tokenization, in a nutshell, refers to using a token (which is basically a virtual currency) to represent ownership of an asset; this token can then be transferred from one user to the next (Chen, Gu et al., 2018).

Tokenization of assets is a fairly new concept and grey literature was very helpful in guiding this research in order to find data/ literature that supported our research questions. The aim of this research was to identify critical success factors of asset tokenization on blockchain and to derive a model that will lead to global adoption. The sub-questions we looked at include, what are the important aspects to consider for asset tokenization; and what other concepts can allude to how asset tokenization should work? These sub-questions aided in deriving a model that may lead to global adoption of tokenization.

Methodology

The goal of the research is to find critical factors that lead to tokenization of assets (digital or real-world) on blockchain technology. To achieve this, a thematic analysis approach will be taken to uncover these factors from the literature collected. Thematic analysis is considered as the most appropriate to allow us to take an interpretivist view of data (Alhojailan, 2012). Tokenization of assets on blockchain is still, in a way, in its nascent stages therefore, analysing grey literature is also valuable (instrumental) in the discovery of these critical success factors.

The data that will be examined is from all the literature that was collected (reviewed) and was deemed relevant to the aim of this research, i.e., literature related to the tokenization of assets on blockchain technology. As the main question or goal, this research paper is guided by the question: What are the critical success factors of asset tokenization on blockchain? -. The paper also takes a look at – as sub-questions:

- ✓ What are the important aspects to consider for asset tokenization?
- ✓ What other concepts can allude to how asset tokenization should work?

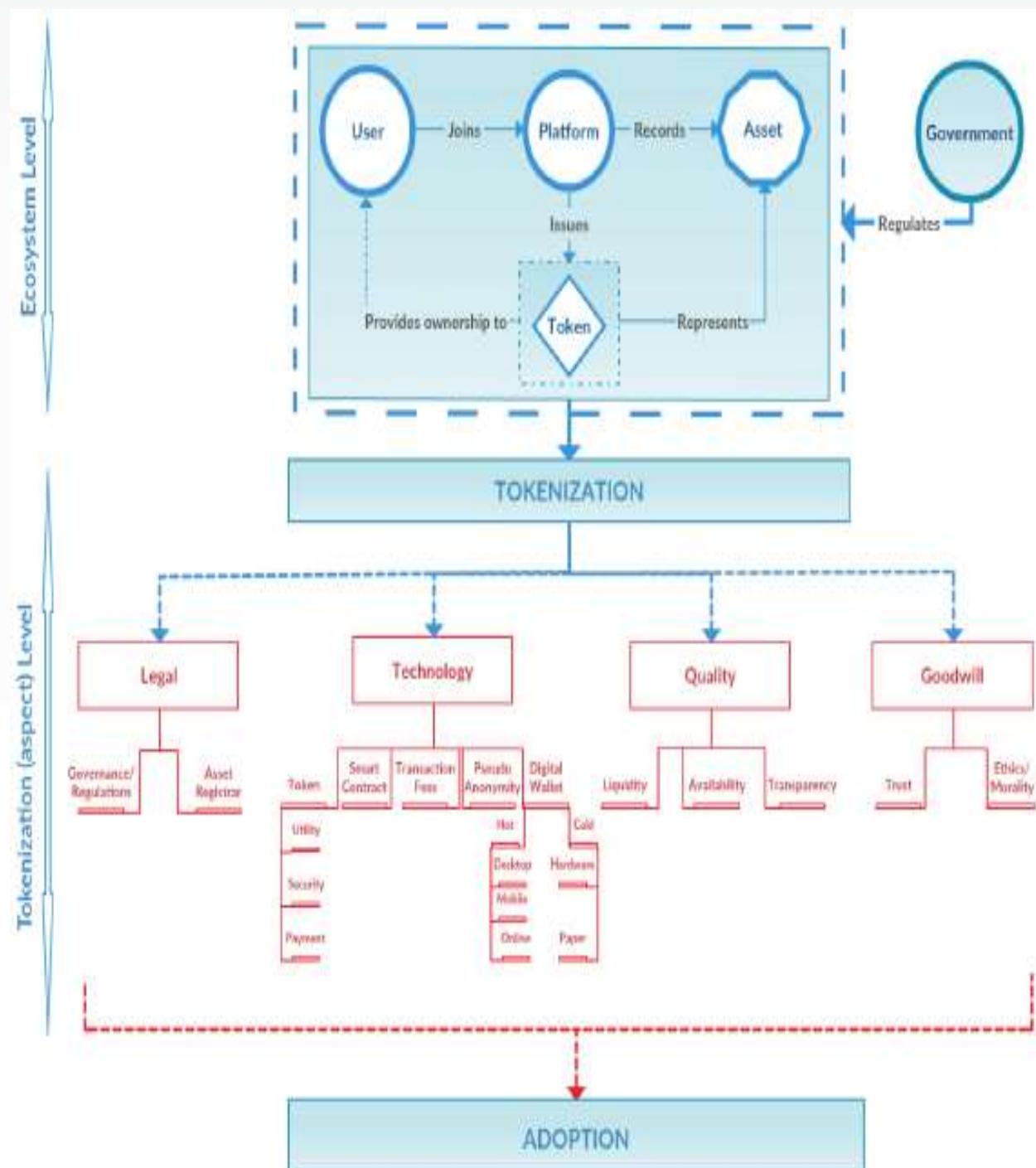
At this stage, the coding of data begins to come up with initial codes that break down the collection into smaller chunks that are more understandable (meaningful). There are multiple methods of coding that are dependent on the type of perspective and research questions (Maguire & Delahunt, 2017). Thus, we will be taking an interpretivist view to analyse the data, and the codes will be guided by the aim of finding critical success factors of tokenization of assets on blockchain; in other words, the research will be leaned towards theoretical thematic analysis (deductive).

The initial codes that will be created will be aligned with or answer the research question, therefore not all text will be coded; because the analysis will aim to produce initial codes that are directly relevant to the research question. A software application called QDA Miner Lite will be used for this coding process or thematic analysis; it allows better management of codes and the analysis as compared to coding manually or by hand.

Upon storing and organizing data in databases to form data warehouses, the amount of data that seemingly becomes overwhelming undergoes the data mining process. Indeed, data mining is a case in which the information is analyzed and summarized with the aim of producing meaningful information. Sophisticated tools of data analysis are employed in a quest to discover relationships and patterns in voluminous datasets. Hence, the tools stretch beyond basic queries and summaries to employ sophisticated algorithms. In a practical illustration of the data mining process, an example could be the case of an online retailer dealing in a number of products. Given that such a business operator could sell many products to a significant number of customers in a typical day, business improvement requires the leveraging of the resultant data. Indeed, it is at this point that the data mining technique comes in to give an insight into products that are

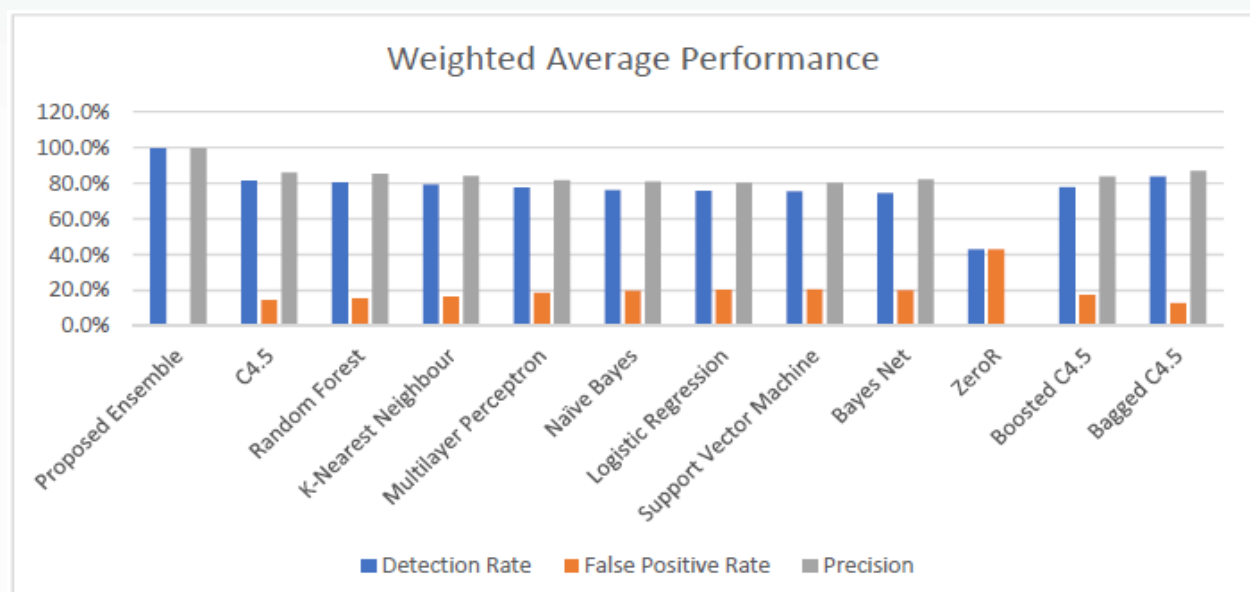
often bought together. In turn, attractive product bundles are created. An additional application of data mining in such a case entails the development of customer profiles. Based on the past purchases, the retailer could employ data mining to predict the products that the customer is likely to seek in future. One of the features of a data warehouse is that it is subject oriented. This feature implies that the warehouse is meant to provide information regarding a given subject and not the ongoing operations of an organization. The subjects include attributes such as revenues, sales, suppliers, customers, and products. Therefore, the focus of data warehousing processes does not lie in the ongoing operations. Instead, the focus lies in information analysis and modeling to support informed decision-making processes. Another feature of data warehouses is that they are integrated. The integration of information characterizes the procedure of constructing data warehouses, employing heterogeneous sources that include flat files and relational databases.

Results and Discussion



Themes and Codes Legal Aspects	Explained
• Governance and Regulations	The platform used should be governed by rules of conduct set to guide the process of tokenization; and guidelines for it manage the process to address various issues such as handling of disputes and determining the jurisdictions in charge in such cases i.e. the platform's or the users' jurisdiction.
• Asset registrar	External party that registers and keeps track of assets ensuring they are not sold on secondary markets (e.g., bid or buy, eBay, offline etc.); while they are still part of the network or still tied to a token (User) on the blockchain platform.
Technology Aspects	
• Token	Tokens are classified as payment, utility and security; security tokens - are subject to securities laws and are deemed appropriate for representation of assets and / or ownership.
• Smart contract	These are programs that self-execute based on conditions set and met by the parties involved and are deemed useful in facilitating transactions without the involvement of intermediaries
• Pseudo-anonymity	The parties who are involved in the transaction may choose to conceal their true / real-world identities and exist on the blockchain platform through public addresses (encryption mechanism), which say nothing about their real-world Identities.
• Transaction fees	These allow remuneration for facilitating every single transaction; also keeps the platform up and running indefinitely. Standardized fees will make the platform uniform while also ensuring every transaction is given the same priority.
• Digital wallet	These wallets hold and secure tokens and are categorized as either hot or cold the former being online and the latter being offline and the most secure.
Quality Aspects	
• Liquidity	There should be an easy and bidirectional flow of token to real money conversion, 24/7; this will increase liquidity (i.e., more investors are likely to be attracted to such an asset market).
• Availability	The blockchain platform used for tokenization should be available worldwide, 24/7, and allow transaction to happen in real time.
• Transparency	Users, or at least their digital identities (public addresses), should be clearly shown to the rest of the network (stakeholders); for instance, show which asset serial number or (public address) is linked to a user's token. <i>Transparency is currently offered by blockchain.</i>
Goodwill Aspects	
• Ethics/ Morality	To increase the level of decentralization, the users and blockchain platform used, should have ethical and moral values that discourage fraudulent operations that would take place during transactions (e.g., the platform conning the users out of their assets or users attempting to exchange non- existent assets).
• Trust	Users have to place trust in the platform's tokens to represent the true value and ownership of assets; in other words, the users have to trust that the asset truly exists - either in the real-world or digitally - that is represented by the token and are not just victims of malpractice (e.g., fraud)

Classification Technique	Accuracy (%)	Detection Rate (%)	False Positive Rate (%)	Precision (%)
C4.5	81.5	81.5	14.6	85.8
Random Forest	80.5	80.5	15.5	85.2
KNN	79.4	79.4	16.5	84.1
MLP	77.7	77.7	18.6	81.7
Naïve Bayes	76.1	76.1	19.8	80.9
Logistic Regression	75.6	75.6	20.3	80.4
SVM	75.4	75.4	20.5	80.2
Bayes Net	74.4	74.4	20	82.2
ZeroR	43.1	43.1	43.1	-
Boosted C4.5	77.9	77.9	17.5	83.8
Bagged C4.5	83.7	83.7	13	87
Proposed Ensemble	99.5	99.5	0.6	99.5



This feature of integration in data warehousing aids in steering effective analysis of the resultant information. Data warehousing is also time variant. The data is linked to certain or particular time-periods; implying that the historical point of view characterizes this provision of information. Lastly, it has been affirmed that data warehousing is non-volatile. This feature implies that as new data is added to the warehouses, the previous information is not erased. Given that the data warehouse stays separate from operational databases, some of the frequent changes that could be made to an organization's operational database are not reflected in data warehouses, implying that an addition of new data does not alter the existing information; making the process non-volatile.

Conclusion

The end goal of this research was to come up with critical success factors through gathering and studying the literature available on asset tokenization. The goal was achieved by a thematic analysis performed on the literature found which led to ideal CSFs that should be considered for successful asset tokenization. A model of asset tokenization was also created, based on our findings, to give us an idea of what elements/ roles are involved in the process. The discussion has managed to tie the research questions together and five critical success factors of asset tokenization were derived from our findings. These factors are very important for the tokenization process to work and for mass (global) adoption. The CSFs were built around the different aspects of the tokenization process and these factors need to be addressed for the process to be successful and assets to be exchanged across the world.

However, the limitations of this research are around the lack of research papers on tokenization; and the use of grey literature – the reliability is often questionable - as guidance towards finding relevant articles/ journals (data) that can assist in answering the research questions and coming up with critical success factors of asset tokenization. Therefore, further research on asset tokenization on blockchain is imperative to uncover some of the aspects that this research paper failed to discover. In theory, asset tokenization seems to be possible and tokens are an ideal way of representing assets; the asset doesn't necessarily need to be in the possession of token holder (owner); these tokens may be used to represent and transfer ownership from one individual to the next.

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